

Security incident report

Section 1: Identify the network protocol involved in the incident

- The primary transport layer protocol involved in the incident is TCP. Additionally is utilized on port 53 for external DNS resolution prior to web session initiation.

Section 2: Document the incident

An unauthorized intrusion occurred on the yummyrecipesforme.com production web server via a brute force attack executed by a disgruntled former employee. Due to the absence of account lockout thresholds and the reliance on factory default administrator credentials, the threat actor successfully guessed the admin password and subsequently modified administrative credentials to lock out the genuine owner. Upon gaining privileged access, the attacker tampered with the web application source code by injecting a malicious JavaScript function. This script served drive by download prompts masquerading as browser updates to site visitors. Once executed, the malicious executable initiated an automated redirection script that hijacked visitor browser sessions to an external spoofed domain, greatrecipiesforme.com, causing system slowdowns and malware infections on client endpoints. Dynamic sandbox analysis utilizing tcmdump confirmed the full attack lifecycle: Initial DNS query via UDP port 53 resolving yummyrecipesforme.com to 203.0.113.22. Standard TCP three-way handshake establishing an HTTP connection on port 80. Execution of the dropped file, triggering secondary DNS resolution for greatrecipiesforme.com. Outbound TCP/HTTP session directed to the spoofed malicious host to deliver secondary malware stages.

Section 3: Recommend one remediation for brute force attacks

- Implement an Account Lockout policy that automatically locks or disables administrative accounts after a specific number of consecutive failed authentication attempts.